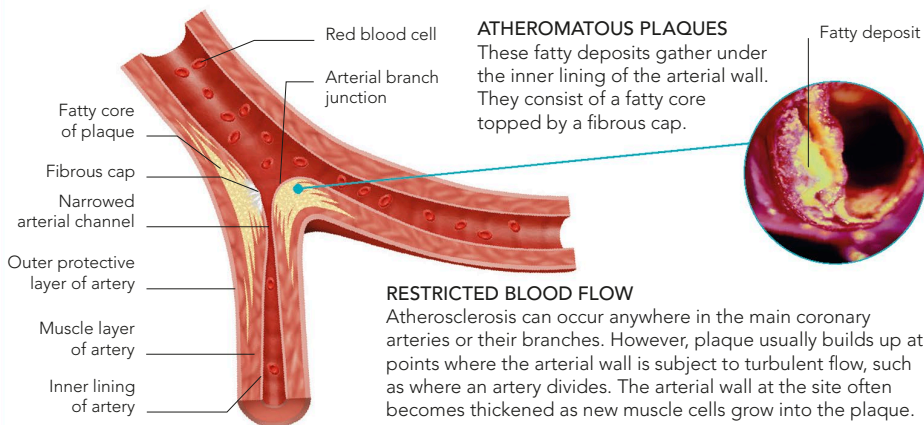


ATHEROSCLEROSIS

The process that leads to atherosclerosis begins with abnormally high levels of excess fats and cholesterol in the blood. These substances infiltrate the lining of arteries, forming deposits known as atheroma. This can happen in any of the body's arteries, including those supplying the brain with blood, when the result may be a stroke. The atheromatous deposits gradually form raised patches known as plaques. These consist of

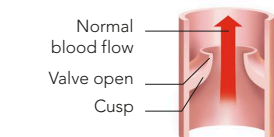
fatty cores within the arterial wall, covered by fibrous caps. The plaques narrow the space within the artery, restricting the overall flow of blood to tissues beyond the site. They also cause turbulence that disrupts the flow of blood, and the eddies over the plaque surface make the blood more likely to clot. The major risk factors for atherosclerosis include smoking, a diet high in saturated fats, lack of exercise, and excess weight.



VALVE DISORDERS

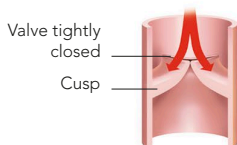
There are two main types of heart valve disorder: stenosis and incompetence.

In stenosis, the valve outlet is too narrow and so restricts blood flow. The condition may be congenital (present at birth), due to an infection such as rheumatic fever, or part of the ageing process. In incompetence, the heart valve does not close fully, allowing backflow of blood. This problem can result from a heart attack or an infection of the valve.



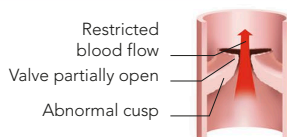
NORMAL VALVE OPEN

As the heart contracts, the high pressure pushes the cusps of the valve open, allowing blood to flow.



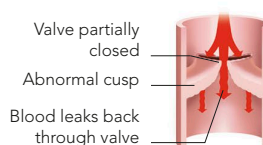
NORMAL VALVE CLOSED

The pressure on the other side of the valve increases and the valve cusps snap shut, preventing backflow.



STENOSIS

The valve stiffens and cannot open fully. Blood flow is restricted, so the heart beats harder to compensate.



INCOMPETENCE

The cusps do not close properly, and blood leaks backward. The heart must work harder to circulate blood.